

Unified Coupler

David Wong

1. Introduction

WRF-CMAQ coupled model was created in 2008 and data communication between the two models, WRF and CMAQ, was done by IOAPI3 BUFFERED file. Later in 2018, the coupled model was expanded to global scale by combining MPAS and CMAQ. In the MPAS-CMAQ coupled model, data communication was done using an ad hoc piece of software developed by David Wong. Construction of an unified coupler is inevitable for fulfilling requirements of software reusability and interoperability.

The unified coupler, which provides a channel for exchanging information/data between met model, either WRF or MPAS, and CMAQ, is one of the key components of the coupled models WRF-CMAQ and MPAS-CMAQ. Figure 1 provides a schematic diagram describing the function of the unified coupler.

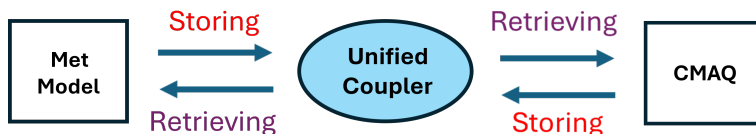


Figure 1: A synopsis of the unified coupler

2. Design Overview

In general, a variable in the met model (Fig. 1) is sent to and stored in the unified coupler, and later on, it will be retrieved and used in CMAQ. This is considered the standard data flow. This data flow pattern holds in the reverse direction, namely, information from CMAQ back to the met

model. In addition, the unified coupler was also designed to relax this data flow paradigm by allowing data structure for communication between models within the coupler be accessed directly. A set of indices were created beforehand to facilitate such access (the list is given at the end of this section). The advantage of this feature is to mitigate potential large memory footprint scenarios.

Meteorological information from WRF is stored in CMAQ based on three classifications: time dependency (time dependent and time independent), dimensionality (2D and 3D), and grid type (cross and dot). MPAS model runs on a horizontal unstructured mesh and mainly data structure is either 1D or 2D, where the first dimension is the number of layers. Furthermore, within CMAQ model data is typically defined as (NCOLS, NROWS, NLAYS) so in the MPAS-CMAQ coupled model, NCOLS is equivalent to the number of mesh points and NROWS is set to 1. For the sake of harmonization, MPAS 1D array is stored in a 2D array (NROWS = 1), and 2D array which is defined as (NLAYS, NCOLS), is converted to (NCOLS, NROWS, NLAYS) and stored in a 3D array. Summarily, in the unified coupler, there are six variables for storage (listed in Table 1 and their names are based on the the naming convention given in Table 2.

Table 1: Six storage variables in the unified coupler.

```
coupler_m2a_ir_2d_data (:,:,)
coupler_m2a_ie_2d_data (:,:,)
coupler_m2a_dr_2d_data (:,:,)
coupler_m2a_dr_3d_data (:,:,)
coupler_m2a_de_3d_data (:,:,)
coupler_a2m_3d_data (:,:,)
```

Table 2: Storage variables naming convention.

category	description
time dependency (i / d)	time independent vs time dependent
dimensionality (2 / 3)	variable dimension
grid type (r / e)	regular cross vs extended dot grid
transfer direction (m2a / a2m)	from met to CMAQ or CMAQ to met

In these six storage variables (Table 1), the third dimension in the 2D and

fourth dimension in the 3D variables are the number of "species" and the fifth dimension with the declaration, 0:1, serves as a circular buffer for data interpolation purposes.

All the species names for m2a and a2m data are sorted alphabetically and stored in the coupler_m2a_vname, and coupler_a2m_vname, respectively. For MPAS-CMAQ coupled model, transport function is performed within MPAS. Hence all CMAQ CGRID transport species are being moved from CMAQ to MPAS to go through transport processes and moving back to CMAQ when it is done. For simplicity, the names of these transport species are not stored. In addition, their values are stored in coupler_m2a_dr_3d_data for both transfer directions further reducing memory footprint.

The unified coupler uses coupler_m2a_nvars, coupler_a2m_nvars, and coupler_m2a_ncvars to keep track numbers of m2a species, numbers of a2m species, and number of CGRID transport species for MPAS-CMAQ coupled model, respectively for bookkeeping purposes.

The following is the list of indices that the unified coupler is used to access specific portion of coupler_m2a* data structure directly:

zf_loc	zh_loc	cfrac3d_loc	eddy_loc	wspd_loc
temp_loc	dens_loc	qv_loc	pres_loc	qc_loc
qr_loc	qg_loc	qi_loc	qs_loc	qc_cu_loc
qi_cu_loc	densa_j_loc	ustar_loc	hfx_loc	pbl_loc

3. Situational Modification

The unified coupler available in the CMAQ release repo supports the current design of the WRF-CMAQ and MPAS-CMAQ coupled models, namely, it provides the appropriate met information either from WRF or MPAS driving the CMAQ model as well as feedback aerosol information back to the met model impacting the radiation calculations. The usage of the unified coupler on WRF-CMAQ and MPAS-CMAQ are tested thoroughly. All the required met variables as well as feedback variables are listed in the coupler_mod.F90 file.

If for some reasons, a user wants to have additional variables from the met model or on the feedback side, here are some guidelines for such modification.

WRF-CMAQ coupled model:

- determine the nature (time dependency, dimensionality, and grid type) met variable and add the new variable in the corresponding variable list and adjust the associated size variable within `coupler_mod.F90` in the WRF-CMAQ section.
- add an appropriate `coupler_data_storing` call in the `twoway_aqprep.F90` with respect to the new variable.
- add an `coupler_data_retrieving` subroutine call in CMAQ receiving the new variable.
- for feedback type variable, add the new feedback variable in the corresponding feedback variable list and adjust the associated size variable within `coupler_mod.F90`.
- modify CMAQ accordingly, in particular the `feedback_setup` subroutine, to formulate this new feedback variable.
- call `coupler_data_storing` with the new variable in `feedback_write` subroutine.
- call `coupler_data_retrieving` with the new variable in `feedback_read` subroutine.

MPAS-CMAQ coupled model:

- determine the dimensionality of the met variable and add the new variable in the corresponding variable list and adjust the associated size variable within `coupler_mod.F90` in the MPAS-CMAQ section.
- add an appropriate `coupler_data_storing` call in the `mpas_cmaq_coupler` subroutine in `mpas_atmchem_interface.F90`.
- add an `coupler_data_retrieving` subroutine call in CMAQ receiving the new variable.
- the direct aerosol SW feedback feature will be added to the MPAS-CMAQ soon.